# COASTAL REFINERY UNIT

## EQUIPMENT INSPECTION REPORT

Report No: IR/CDU/2026/0417 Date of Inspection: 28 September 2026

Equipment: Heat Exchanger E-204 (Crude Preheat Train, CDU-1)

Type: Shell and Tube, 420 tubes Design Pressure: 14.0 kg/cm2

Inspected by: R. Menon, Senior Inspection Engineer

### FINDINGS

1. Shell thickness: nominal 12.0 mm. Minimum measured 9.2 mm at CML-07. Minimum required thickness as per data sheet is 8.5 mm.
2. Tube bundle: 14 of 420 tubes found leaking and plugged (3.3 percent).
3. Channel head flange gasket found leaking and hardened.
4. Corrosion under insulation at nozzle N3: pitting up to 0.6 mm deep.
5. Hydrostatic test at 18.2 kg/cm2 held for 30 minutes. No pressure drop, no visible leak. Test passed.

### RECOMMENDATION

Replace channel head gasket. Clean, recoat and re-insulate nozzle N3. Exchanger may be returned to service after the above repairs. Estimated repair cost: Rs 4.8 lakh (budget head: CDU-1 maintenance).

Signature: R. Menon Reviewed: S. Kulkarni, Manager (Inspection)

INSPECTION DEPT